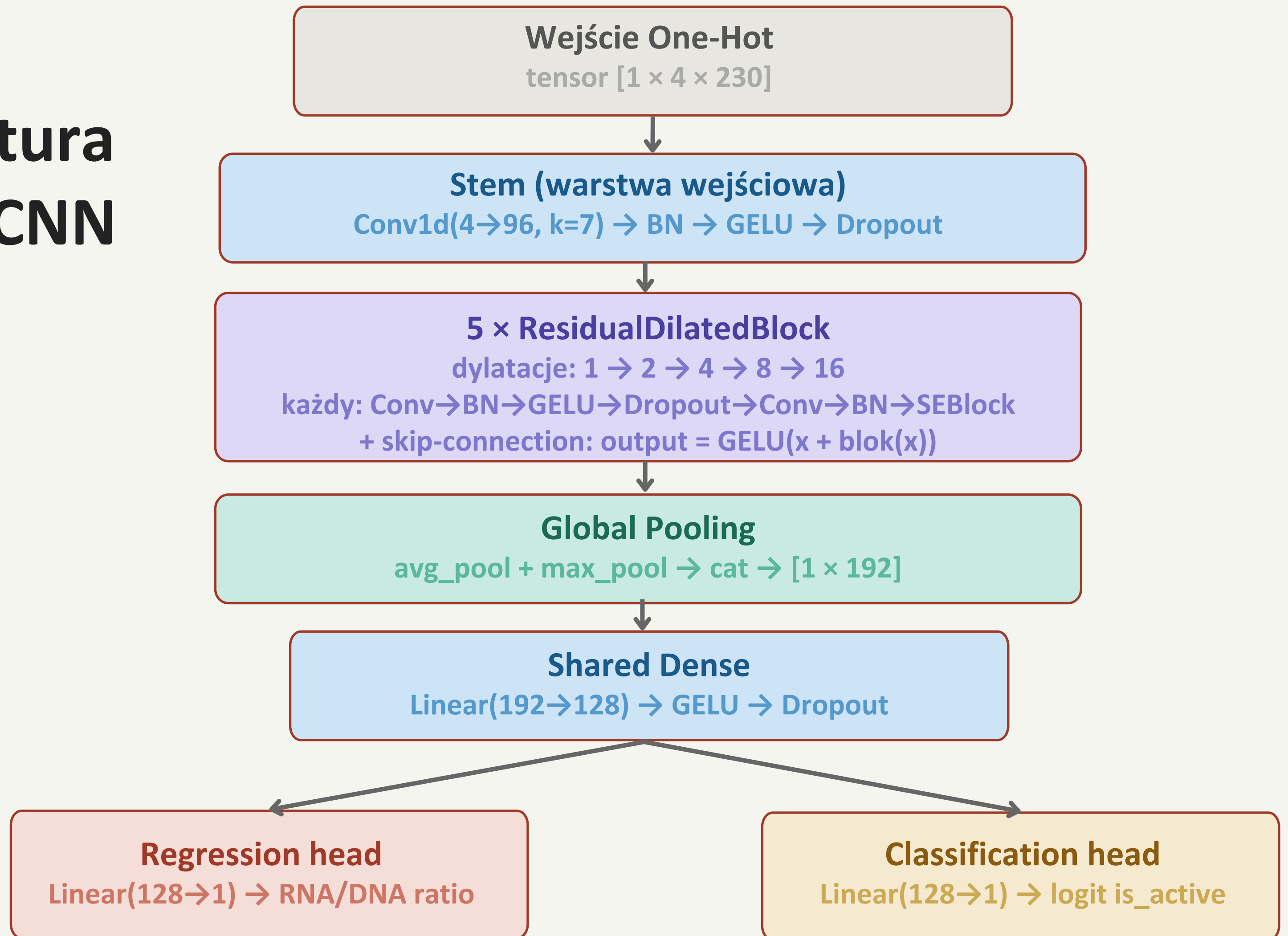


Final Group Project:

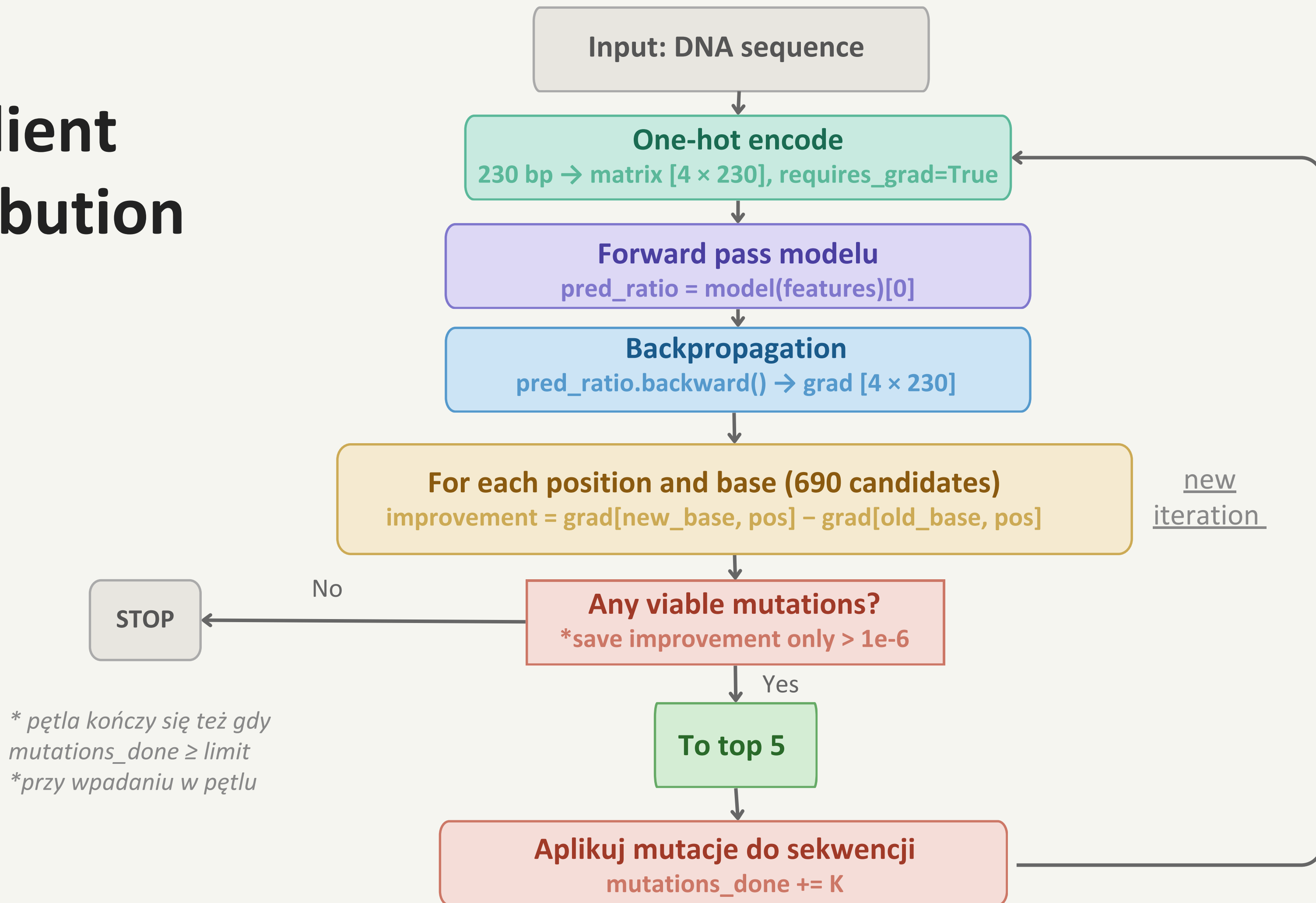
Deep Learning for Regulatory DNA Sequence Optimization

Michał Zgieb
Maciej Kowalski
Katsiaryna Dubrouskaya

Architektura modelu CNN



Gradient Attribution



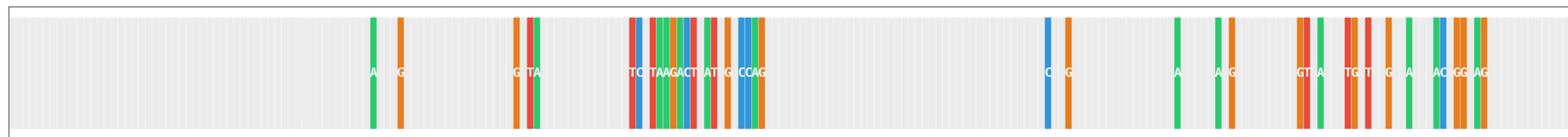
Results

Sequence	Original Activity (approx.)	Introduced Mutations	Achieved activity
A. Seq 1	~2.5	40	4.38
A. Seq 2	~2.7	16	3.67
A. Seq 3	~1.1	20	1.92
B. Seq	---	80	4.98

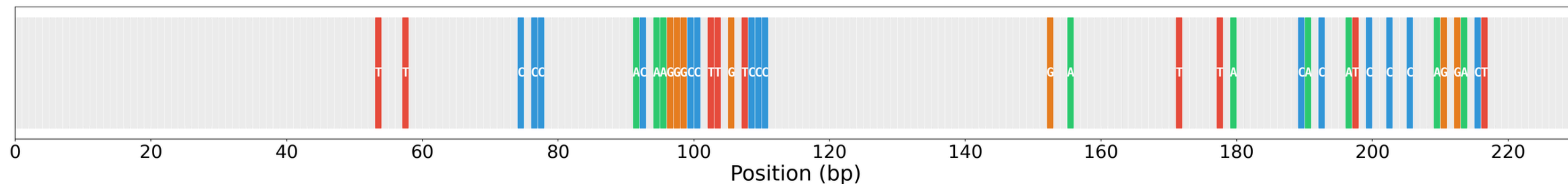
Sequence comparison — A_seq1_broken (40 mutated bp along 230 bp)

A C G T unchanged

Original



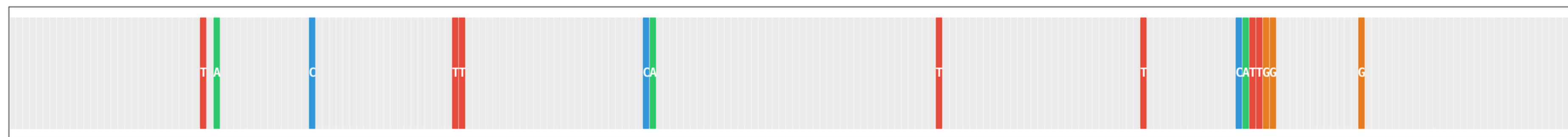
Mutated



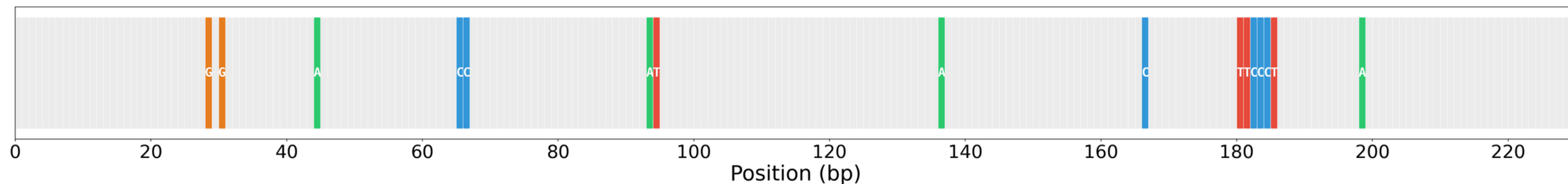
Sequence comparison — A_seq2_broken (16 mutated bp along 230 bp)

A C G T unchanged

Original



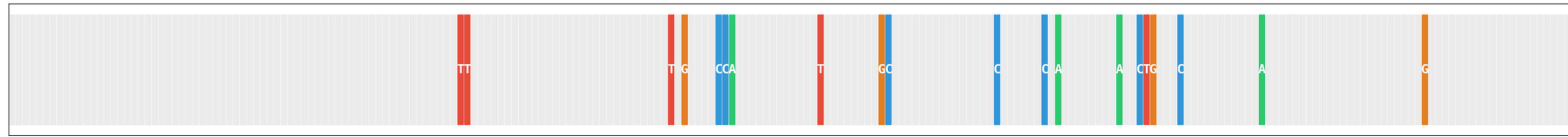
Mutated



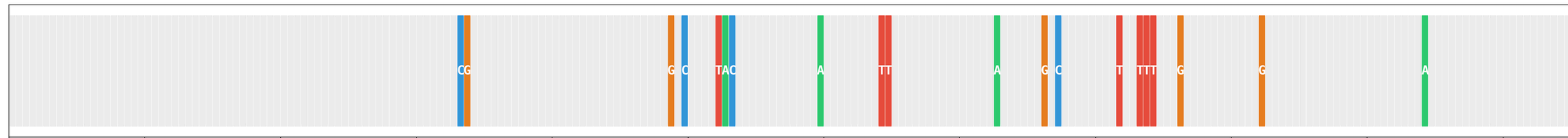
Sequence comparison — A_seq_3_broken (20 mutated bp along 230 bp)

A C G T unchanged

Original



Mutated

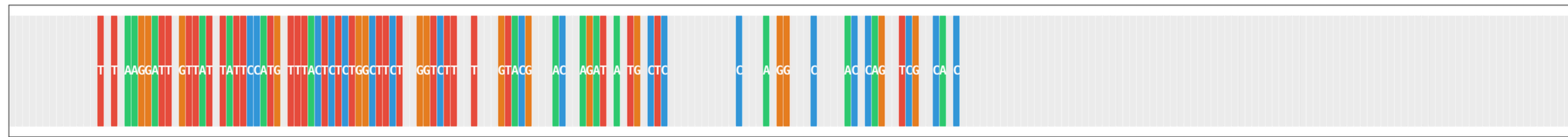


Position (bp)

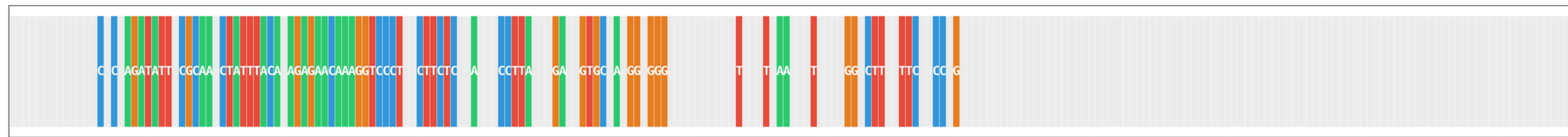
Sequence comparison — B_seq_4_no_active (80 mutated bp along 230 bp)

A C G T unchanged

Original



Mutated



Position (bp)